

Supplementary Material for: *Evolutionary Optimisation of Data Augmentation Policies for Medical Image Classification*

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Overview

This document provides supplementary material for the paper “*Evolutionary Optimisation of Data Augmentation Policies for Medical Image Classification*”. It includes additional experimental details, extended results, and reproducibility information that could not be included in the main paper due to space constraints.

1 Additional Experimental Details

1.1 Datasets Details

We utilised four subsets of the MedMNIST v2 benchmark [?], a large-scale collection of standardised biomedical images. The datasets were selected to cover diverse imaging modalities (Ultrasound, Dermatoscopy, X-Ray, and CT) and task types (Binary and Multi-class classification).

Source and Licensing: All datasets are available at the official MedMNIST repository: <https://medmnist.com/>. The MedMNIST dataset is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0), except DermaMNIST under Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0).

Preprocessing:

- **Resolution:** All images were used in their provided resolution of 28×28 pixels for the search phase. For the transferability experiments, images were upscaled to 224×224 using bicubic interpolation.
- **Normalisation :** Input images were normalised to a standard range by subtracting a mean of 0.5 and dividing by a standard deviation of 0.5 for all channels ($\mu = [0.5, 0.5, 0.5]$, $\sigma = [0.5, 0.5, 0.5]$). For grayscale images (BreastMNIST, PneumoniaMNIST, OrganCMNIST), channels were replicated to match the 3-channel input expectation of ResNet models before normalisation.

- **Splits:** We strictly adhered to the official training, validation, and testing splits provided by MedMNIST v2 to ensure fair comparison with baselines.

Table 1 summarises the sample distribution for each selected dataset.

Table 1: Overview of the selected MedMNIST v2 datasets.

Dataset	Task	Samples			Classes
		Train	Val	Test	
BreastMNIST	Binary	546	78	156	2
DermaMNIST	Multi-class	7,007	1,003	2,005	7
PneumoniaMNIST	Binary	4,708	524	624	2
OrganCMNIST	Multi-class	13,000	2,392	8,268	11

1.2 Model Architectures

We used standard ResNet-18 and ResNet-50 architectures. The models were trained from scratch without pre-trained weights. To adapt them to the datasets, we applied the following configurations:

- **Output Layer:** The final fully connected layer was replaced to match the specific number of classes (C) of each dataset.
- **Input Adaptation:** As standard ResNets expect 3-channel inputs, grayscale images were processed by replicating the single channel across the RGB dimensions.
- **Initialisation :** All models utilised the default PyTorch initialisation, which uses the He (Kaiming Normal) scheme for convolutional layers.

2 Additional Figures

Figures 1 and 2 show the frequency of augmentation operations present in the final evolved populations for BreastMNIST. Figure 1 shows that ResNet-18 frequently selects geometric transformations (e.g., VerticalFlip, GridDistortion) and edge enhancement (Sharpen). This suggests the shallower network benefits from operations that preserve structural information. In contrast, Figure 2 reveals that ResNet-50 prioritises intensity distortions such as ImageCompression and illumination changes. We hypothesise this occurs because the deeper architecture requires stronger regularisation to prevent memorising the small training set. The evolutionary search adapts by selecting augmentations that force the model to learn more robust, texture-invariant features.

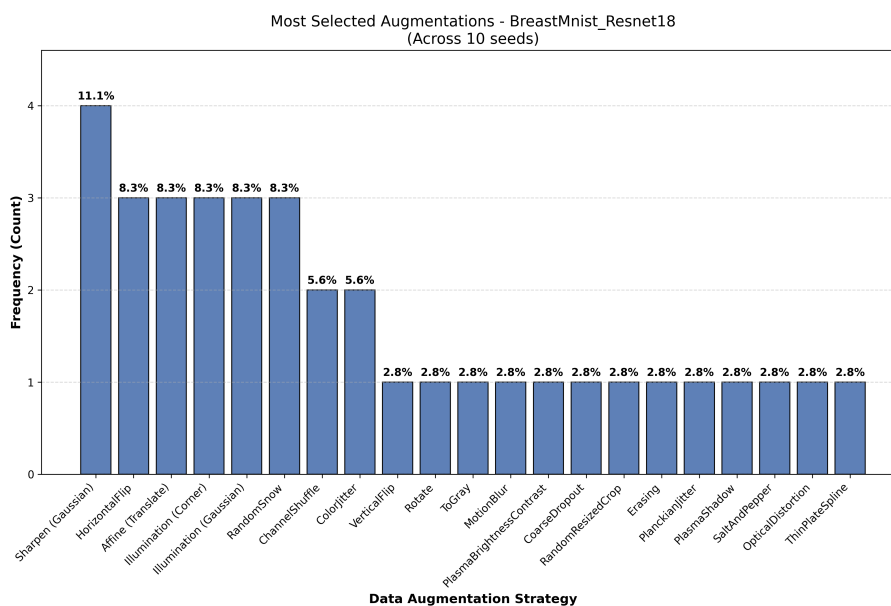


Fig. 1: Most frequent augmentations for ResNet-18.

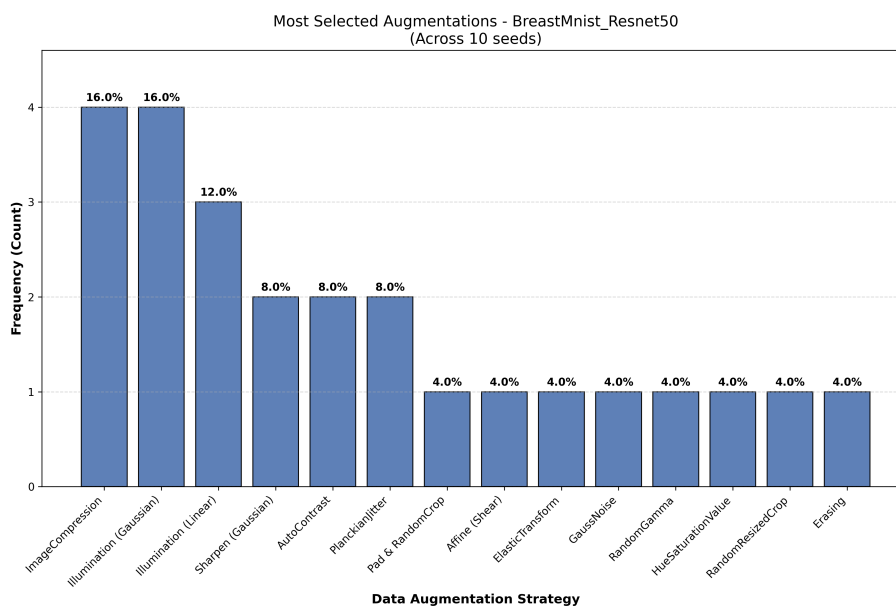


Fig. 2: Most frequent augmentations for ResNet-50.

2.1 Best evolved Individuals

Tables 2 and 3 detail the exact phenotypes of the highest-performing individuals discovered by DAOP. For each dataset and architecture, we report the complete sequence of augmentation operations, their probability of application (p), and the four raw magnitude parameters (m_1 to m_4). These raw values, ranging within $[0, 1]$, correspond to the operation-specific intensity ranges defined in the methodology. Providing these specific configurations enables full reproducibility of our best-reported results.

Table 2: **ResNet-18: Detailed phenotypes of the best-evolved policies.** Description includes operation sequence, probability (p), and raw parameters ($[m_1..m_4]$).

Dataset	Best Evolved Policy (ResNet-18)
BreastMNIST	ColorJitter ($p = 0.47$) [0.44, 0.12, 0.06, 0.27]
	VerticalFlip ($p = 0.81$) [0.92, 0.58, 0.69, 0.06]
DermaMNIST	Illumination (Corner) ($p = 0.71$) [0.32, 0.72, 0.60, 0.97]
	HorizontalFlip ($p = 0.32$) [0.59, 0.55, 0.01, 0.82]
	Perspective ($p = 0.97$) [0.22, 0.37, 0.00, 0.70]
PneumoniaMNIST	RandomSnow ($p = 0.09$) [0.69, 0.74, 0.20, 0.86]
	PlasmaBrightContrast ($p = 0.04$) [0.67, 0.10, 0.45, 0.37]
	Illumination (Gaussian) ($p = 0.06$) [0.62, 0.19, 0.62, 0.26]
	Pad & Crop ($p = 0.86$) [0.62, 0.17, 0.21, 0.86]
	ChannelShuffle ($p = 0.54$) [0.66, 0.17, 0.23, 0.07]
OrganCMNIST	SaltAndPepper ($p = 0.66$) [0.23, 0.00, 0.04, 0.72]
	Affine (Translate) ($p = 0.51$) [0.70, 0.25, 0.03, 0.33]
	ThinPlateSpline ($p = 0.25$) [0.16, 0.62, 0.72, 0.00]

3 Statistical Tests

To assess whether DAOP yields statistically significant improvements over baseline and AutoML methods, we performed one-sample superiority tests based on repeated DAOP runs.

- **Type of test.** For each dataset and metric (AUC and ACC), we compared the 10 independent DAOP runs against each fixed reference method (Baseline ResNet-18, Baseline ResNet-50, Auto-sklearn, AutoKeras, and Google AutoML) using one-sample, one-sided hypothesis tests. Normality of the DAOP run distributions was assessed using the Shapiro–Wilk test. When normality could be assumed ($p \geq 0.05$), we used a one-sample t -test; otherwise, we used the nonparametric Wilcoxon signed-rank test. The alternative hypothesis was $H_1 : \mu_{\text{DAOP}} > \mu_0$, where μ_0 is the fixed reference score.
- **Number of resamples or folds.** Statistical tests were based on $n = 10$ independent DAOP runs (random seeds) per dataset and DAOP variant.

Table 3: **ResNet-50: Detailed phenotypes of the best evolved policies.** Description includes operation sequence, probability (p), and raw parameters ($[m_1..m_4]$).

Dataset	Best Evolved Policy (ResNet-50)
BreastMNIST	Illumination (Gaussian) ($p = 0.49$) [0.52, 0.82, 0.20, 0.19]
	ChannelShuffle ($p = 0.38$) [0.96, 0.48, 0.15, 0.88]
	Affine (Translate) ($p = 0.43$) [0.30, 0.71, 0.23, 0.34]
DermaMNIST	Posterize ($p = 0.46$) [1.00, 0.85, 0.70, 1.00]
	RandomResizedCrop ($p = 0.62$) [0.78, 0.47, 0.00, 0.40]
	VerticalFlip ($p = 0.26$) [0.77, 0.73, 0.33, 0.48]
	Illumination (Gaussian) ($p = 0.59$) [0.64, 0.07, 0.25, 0.08]
	Rotate ($p = 0.32$) [0.99, 0.40, 0.47, 0.49]
PneumoniaMNIST	RandomBrightnessContrast ($p = 0.13$) [0.67, 0.66, 0.14, 0.51]
	Pad & Crop ($p = 0.95$) [0.46, 0.78, 0.23, 0.53]
OrganCMNIST	Illumination (Gaussian) ($p = 0.81$) [0.47, 0.62, 0.85, 0.57]
	Affine (Translate) ($p = 0.87$) [0.42, 0.79, 0.58, 0.36]
	ToGray ($p = 0.97$) [0.71, 0.90, 0.71, 0.18]
	ToGray ($p = 0.00$) [0.89, 0.22, 0.23, 0.06]
	Sharpen (Gaussian) ($p = 0.12$) [0.64, 0.68, 0.69, 0.44]

Table 4: Statistical significance of DAOP Mean (ResNet-18) vs baselines and AutoML. Cells report **p-value;Sig**, where ** indicates significance at 99% ($\alpha = 0.01$) and * indicates significance at 95% ($\alpha = 0.05$) after Holm correction.

Dataset & Metric	Baseline (R18)	Baseline (R50)	Auto-sklearn	AutoKeras	Google AutoML
BreastMnist AUC	0.00009;**	0.00001;**	0.00001;**	0.00001;**	0.00440;**
BreastMnist ACC	0.01740;*	0.00001;**	0.00001;**	0.00001;**	0.01630;*
DermaMnist AUC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00001;**
DermaMnist ACC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00080;**
PneumoniaMnist AUC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	1.00000;
PneumoniaMnist ACC	0.02800;*	1.00000;	0.02800;*	1.00000;	1.00000;
OrganCMnist AUC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00001;**
OrganCMnist ACC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00001;**

No resampling or cross-validation folds were used in the statistical testing phase, as reference methods were available only as single summary values.

- **Significance level α .** We report results at both the 95% ($\alpha = 0.05$) and 99% ($\alpha = 0.01$) confidence levels. Primary conclusions in the main paper are based on the more conservative $\alpha = 0.01$ threshold.
- **Multiple comparison correction.** To control for multiple hypothesis testing across reference methods, the Holm-Bonferroni correction was applied within each dataset–metric–DAOP family (i.e., across the five reference comparisons for a fixed dataset, metric, and DAOP variant). Holm-corrected p -values and boolean significance indicators (True/False) are reported.

DAOP Max results are reported for completeness but are not subjected to statistical testing, as they represent best-of-run values and do not reflect reproducible performance across independent runs.

Table 5: Statistical significance of DAOP Mean (ResNet-50) vs baselines and AutoML. Cells report **p-value;Sig**, where ** indicates significance at 99% ($\alpha = 0.01$) and * indicates significance at 95% ($\alpha = 0.05$) after Holm correction.

Dataset & Metric	Baseline (R18)	Baseline (R50)	Auto-sklearn	AutoKeras	Google AutoML
BreastMnist AUC	0.00020;**	0.00001;**	0.00001;**	0.00001;**	1.00000;
BreastMnist ACC	0.19880;	0.00001;**	0.00001;**	0.00030;**	0.19880;
DermaMnist AUC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00001;**
DermaMnist ACC	0.00001;**	0.00001;**	0.00001;**	0.00040;**	1.00000;
PneumoniaMnist AUC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	1.00000;
PneumoniaMnist ACC	0.00020;**	1.00000;	0.00020;**	0.09480;	1.00000;
OrganCMnist AUC	0.00490;**	0.00490;**	0.00490;**	0.00490;**	0.00490;**
OrganCMnist ACC	0.00001;**	0.00001;**	0.00001;**	0.00001;**	0.00001;**